

ARIJIT BHATTACHARJEE

Bengaluru, India , +91 7005467390 , arijit1201@gmail.com

SUMMARY

Backend platform engineer with 6 years building event-driven, config-driven distributed systems on Java/Spring Boot and Azure - specialising in Kafka-based streaming pipelines, serverless compute, and high-volume data ingestion. Track record of owning platform deliverables end to end: designed integration layers reducing onboarding from weeks to days, built and load-tested event ingestion systems at production scale, and led AI-assisted cloud asset analysis across 400+ resources. Currently deepening expertise in LLM toolchain design and agentic systems engineering, applied to both production workflows and a large-scale COBOL-to-Java modernisation initiative.

TECHNICAL SKILLS

- **Languages:** Java (primary), Python, SQL, COBOL (analysis-level), JavaScript
- **AI & LLM Tooling:** GitHub Copilot, Azure OpenAI, Gemini, Ollama, LMStudio, RAG, context window management, prompt engineering
- **Frameworks:** Spring Boot, Spring, GraphQL (Apollo Server), JUnit, FastAPI, Flask
- **Distributed Systems:** Apache Kafka (batch/parallel consumption, load testing), Azure Event Hubs, event-driven architecture, config-driven transformation (JOLT, JSON Schema)
- **Databases:** Cosmos DB (NoSQL), MongoDB, MySQL, Redis
- **Cloud & Infrastructure:** Azure (Functions, Logic Apps, Blob, APIM, AKS, OpenAI, Resource Graph, Monitor), Docker, Kubernetes
- **DevOps & Quality:** Jenkins, Groovy, GitHub Actions, AccelQ, SonarQube, CI/CD, Git

PROFESSIONAL EXPERIENCE

Optum (UnitedHealth Group) , Hyderabad / Bengaluru

Senior Software Engineer

Mar 2024 – Present

- Leading a solo AI-assisted COBOL-to-Java modernisation initiative on a 47K-line PBM benefit-group orchestrator - built Python dependency analysis tooling to map the full program call graph and establish a dependency-ordered migration sequence across 100+ programs; designed a section-level spec extraction pipeline and hexagonal Java PoC with capability mapping and source traceability annotations.
- Architected a configuration-driven event processing layer externalising transformation rules (JOLT), validation schemas (JSON Schema), and routing config into blob storage - eliminating per-integration code duplication and decoupling onboarding from the release cycle; reduced new integration onboarding from 2-4 weeks of custom development to 3-4 days of config specification, with ongoing mid-sprint payload changes now handled via config updates rather than code deployments.
- Designed and owned Kafka-to-serverless ingestion pipelines for polymorphic, high-throughput payloads - covering fault-tolerance strategy, dead-letter handling, and consumer performance tuning; built a Python load testing framework to validate throughput, lag behaviour, and failure recovery under production-scale load.
- Led AI-assisted analysis of 400+ legacy cloud assets using Azure Resource Graph and Azure OpenAI - automated business logic extraction and classification at scale, producing a structured decommissioning roadmap and saving hundreds of hours of manual engineering effort; resolved critical production security vulnerabilities and drove cloud cost optimisation through systematic serverless resource decommissioning.
- Identified consumer throughput and NoSQL read latency as performance bottlenecks in high-volume asynchronous workflows; introduced Kafka batch consumption with parallel processing and a Redis caching layer over Cosmos DB for frequently accessed reference data - measurably improving consumer throughput and reducing per-event processing latency; developed GraphQL APIs enabling flexible, field-level data access for internal platform consumers.

Optum (UnitedHealth Group) , Hyderabad

Software Engineer

Jul 2020 – Mar 2024

- Architected reusable, configuration-driven serverless components consumed across multiple platform orchestrations - contributed to \$600K+ in annual savings through automated resource orchestration and reduced duplication.
- Delivered 3 platform innovations adopted organisation-wide (~\$200K estimated impact each): a global reusable events API, a config-driven event mapping function, and a config-driven NoSQL persistence utility - standardising shared engineering patterns across teams.
- Developed Java/Spring Boot backend APIs spanning member preferences, notifications, scheduling, medication, and pricing workflows - integrating multiple enterprise downstream systems into stable, well-tested API contracts.
- Established CI/CD pipelines in Jenkins and GitHub Actions with end-to-end automated testing for asynchronous workflows - sustaining 90%+ code coverage and Quality Gate 8+ compliance as enforced engineering quality standards across the team.
- Improved Kafka consumer services through batch consumption and parallel processing; built Python automation scripts for bulk data operations and large-scale failure analysis across tens of thousands of records; led migration of workflow orchestrations to a standardised runtime and owned production incident resolution on high-visibility platform releases.

TECHNICAL PROJECTS

InfernalCV - LLM-Powered Document Analysis

Modular multi-provider LLM inference tool with PDF/DOCX parsing and routing across Gemini, OpenRouter, and Ollama - demonstrates LLM system design, multi-model orchestration, and context-scoped inference across provider backends.

Tech: *Python, FastAPI, PyPDF2, Azure OpenAI, Gemini, Ollama*

EDUCATION

National Institute of Technology Jamshedpur

B.Tech, Computer Science and Engineering

2016 – 2020 , CGPA: 8.80 / 10